MESH-ONLY CONVERGENCE FOR THE CANONICAL WALK ON ORTHOGONAL TILINGS

Abstract. We prove that the geometric alternatives in Bou-Rabee and Gwynne [2] are not needed for convergence of the canonical random walk on an orthogonal tiling. The only geometric assumption is that the mesh tends to zero. The proof uses a continuous-time parametrization whose reversible measure is tile volume. A Euclidean BV–Sobolev inequality yields a defective Nash inequality and hence heat-kernel smoothing after the mesh time scale. A local power potential gives a lower bound for the amount of tile-volume time needed to cross a fixed Euclidean ball. These estimates, together with the finite-volume energy estimate of Bou-Rabee and Gwynne [2, Proposition 2.6], give a direct quantitative bound for the uniform error in the discrete Dirichlet problem.

Draft proof note

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1. Statement of the result

We use the notation of Bou-Rabee and Gwynne [2, Sections 1.1 and 2.1]. Let $D \subset \mathbb{R}^d$ be open. An *orthogonal tiling* of D is a locally finite collection $(P_v)_{v\in\mathcal{V}}$ of closed convex polytopes with pairwise disjoint interiors, together with marked points $v \in \operatorname{int} P_v$, such that the polytopes cover D. Distinct vertices $v, w \in \mathcal{V}$ are joined by an edge when

$$H_e := P_v \cap P_w, \qquad e = \{v, w\},$$

is a $(d-1)$-dimensional facet. We assume that $w - v$ is orthogonal to the affine span of H_e. Write

$$m_v := \operatorname{vol}_d(P_v), \qquad b(e) := \operatorname{vol}_{d-1}(H_e), \qquad \|e\| := \|w-v\|, \qquad a(e) := \frac{b(e)}{\|e\|}. \tag{1}$$

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The weighted adjacency graph is denoted by $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Its mesh is

$$\epsilon := \sup_{v \in \mathcal{V}} \operatorname{diam}(P_v). \tag{2}$$

For $U \Subset D$, let

$$\mathcal{V}[U] := \mathcal{V} \cap U, \qquad \partial\mathcal{V}[U] := \{w \in \mathcal{V} \setminus \mathcal{V}[U] : w \sim v \text{ for some } v \in \mathcal{V}[U]\},$$

and set

$$\overline{\mathcal{V}[U]} := \mathcal{V}[U] \cup \partial\mathcal{V}[U].$$

We write $\overline{\mathcal{E}[U]}$ for the set of edges with at least one endpoint in $\mathcal{V}[U]$, and

$$\widehat{U} := \bigcup_{v \in \overline{\mathcal{V}[U]}} P_v. \tag{3}$$

Local finiteness implies that $\overline{\mathcal{V}[U]}$ is finite whenever U is bounded and $\overline{U} \subset D$.

For a function $f : \mathcal{V} \to \mathbb{R}$, the weighted discrete gradient and Laplacian are

$$\nabla_a^{\mathcal{G}} f(v, w) := a(v, w)(f(w) - f(v)), \tag{4}$$

$$\Delta_a^{\mathcal{G}} f(v) := \sum_{w \sim v} a(v, w)(f(w) - f(v)). \tag{5}$$

If f vanishes outside a finite set, its Dirichlet energy and its weighted L^p norms are

$$\mathcal{D}(f) := \sum_{e=\{v,w\} \in \mathcal{E}} a(e)(f(w) - f(v))^2, \qquad \|f\|_{p,m}^p := \sum_{v \in \mathcal{V}} m_v |f(v)|^p. \tag{6}$$

For a function on $\overline{\mathcal{V}[U]}$ which vanishes on $\partial\mathcal{V}[U]$, we extend it by zero to $\mathcal{V} \setminus \overline{\mathcal{V}[U]}$. With this convention, $\mathcal{D}(f)$ agrees with the energy $\mathcal{D}(f, U)$ of Bou-Rabee and Gwynne [2, Equation (2.7)].

Let h_C be harmonic and C^2 on an open neighborhood of $\overline{U}$. Whenever every connected component of $\mathcal{V}[U]$ is joined to $\partial\mathcal{V}[U]$, its discrete Dirichlet extension is the function $h_D : \overline{\mathcal{V}[U]} \to \mathbb{R}$ satisfying

$$\begin{cases} \Delta_a^{\mathcal{G}} h_D = 0 & \text{on } \mathcal{V}[U], \\ h_D = h_C & \text{on } \partial\mathcal{V}[U]. \end{cases} \tag{7}$$

The following finite-network fact will be used repeatedly.

Lemma (Dirichlet principle and maximum principle). Let A be a finite set of vertices, let ∂A be a finite set of neighboring boundary vertices, and suppose that every connected component of A is joined by an edge path in $A \cup \partial A$ to ∂A. For prescribed boundary values $g : \partial A \to \mathbb{R}$, there is a unique function $u : A \cup \partial A \to \mathbb{R}$ which equals g on ∂A and satisfies $\Delta_a^{\mathcal{G}} u = 0$ on A. Moreover,

$$\min_{\partial A} g \le u(v) \le \max_{\partial A} g, \qquad v \in A \cup \partial A. \tag{8}$$

Proof. On the affine space of functions equal to g on ∂A, minimize

$$\frac{1}{2} \sum_{\{v,w\} :\, \{v,w\} \cap A \neq \varnothing} a(v, w)(u(v) - u(w))^2.$$

A minimizer exists because the space is finite dimensional. The path-to-boundary assumption makes the quadratic part positive definite on functions vanishing on ∂A, so the minimizer is unique. Its Euler–Lagrange equations are $\Delta_a^{\mathcal{G}} u = 0$ on A.

For the maximum principle, suppose that u attains its maximum at an interior vertex v. Harmonicity gives

$$u(v) = \frac{1}{a(v)} \sum_{w \sim v} a(v,w)u(w), \qquad a(v) := \sum_{w \sim v} a(v,w).$$

Since every weight is positive, every neighbor of v must have the same value. Repeating this argument along an edge path to the boundary shows that the maximum is attained on ∂A. Applying the same argument to $-u$ gives the minimum bound. □

We next verify the path-to-boundary condition in the small-mesh regime needed below. Since $\overline{U} \Subset D$, choose $\delta > 0$ such that $B_\delta(\overline{U}) \subset D$. The number of connected components of D which meet $\overline{U}$ is finite: if one chooses a point of $\overline{U}$ in each such component, the corresponding balls of radius $\delta/2$ are pairwise disjoint and lie in the bounded set $B_\delta(\overline{U})$. Denote these components by $D_1, \dots, D_N$. For each j, the compact set $\overline{U} \cap D_j$ is a proper subset of D_j, so we may choose

$$y_j \in D_j \setminus \overline{U}, \qquad \rho_j := \operatorname{dist}(y_j, \overline{U}) > 0.$$

Set $\rho_U := \min_{1 \le j \le N} \rho_j$. Suppose that $\epsilon < \rho_U$, and fix $v \in \mathcal{V}[U]$. Let D_j be the component containing v, and choose a cell P_{w_j} containing y_j. Since $\|w_j - y_j\| \le \operatorname{diam}(P_{w_j}) \le \epsilon < \rho_j$, one has $w_j \notin U$.

A polygonal path in D_j joins v to w_j. Its image is compact and contained in the open set D_j, so it has a compact tubular neighborhood contained in D_j. By local finiteness, only finitely many cells meet this neighborhood. Perturbing the path inside the neighborhood, we may arrange that it avoids every face of dimension at most $d-2$ of these finitely many cells and crosses each encountered facet transversely. Reading the cells in the order in which the perturbed path visits them gives an edge path in the facet-adjacency graph from v to w_j. The first marked point on this edge path which does not belong to U is a vertex of $\partial\mathcal{V}[U]$. Thus every connected component of $\mathcal{V}[U]$ is joined to $\partial\mathcal{V}[U]$ whenever $\epsilon < \rho_U$, and the finite-network lemma above gives existence and uniqueness for (7) in this regime.

Theorem 1.1 (Mesh-only Dirichlet convergence)**.** *Let $d \ge 2$, let $U \Subset D$ be bounded, and let $(\mathcal{G}_n)_{n \ge 1}$ be a sequence of orthogonal tilings of D whose mesh sizes ϵ_n tend to zero. Let h_C be C^2 and harmonic on a neighborhood of $\overline{U}$, and, for all sufficiently large n, let $h_{D,n}$ solve (7) on $\mathcal{G}_n$. Then*

$$\lim_{n \to \infty} \max_{v \in \mathcal{V}_n[U]} |h_{D,n}(v) - h_C(v)| = 0. \tag{9}$$

More precisely, if K is a fixed bounded open set such that $\overline{U} \subset K \Subset D$ and h_C is C^2 and harmonic on a neighborhood of $\overline{K}$, then, for all sufficiently small ϵ_n,

$$\max_{v \in \mathcal{V}_n[U]} |h_{D,n}(v) - h_C(v)| \le C\epsilon_n^{\beta_d}, \qquad \beta_d := \frac{4}{3d+8}, \tag{10}$$

where C depends only on d, U, K, and $\|h_C\|_{C^2(K)}$. In particular, the convergence is uniform over families for which these quantities are uniformly bounded.

The process introduced in section 3 is only an auxiliary clock. Its embedded jump chain has transition probabilities

$$\mathbb{P}[v \to w] = \frac{a(v,w)}{\sum_{u \sim v} a(v,u)},$$

which are exactly those of the canonical discrete-time walk in Bou-Rabee and Gwynne [2, Equation (1.1)]. Changing the holding times therefore does not change harmonic functions, exit locations, or harmonic measure.

Corollary 1.2. *In the orthogonal-tiling setting of Bou-Rabee and Gwynne [2, Theorems A and B], the mesh hypothesis alone verifies Assumption D of Bou-Rabee and Gwynne [2, Section 4]. Consequently, the alternatives (I), (II), and (III) used there to establish Assumption D are unnecessary: the harmonic-measure conclusion of Bou-Rabee and Gwynne [2, Proposition 4.2] and the stopped-walk convergence conclusion of Bou-Rabee and Gwynne [2, Theorem 4.1] hold under the remaining hypotheses.*

2. A defective Nash inequality

Throughout this section, $\mathcal{G}$ is a fixed orthogonal tiling with mesh $\epsilon < \infty$. Every function under consideration has finite support and satisfies

$$\bigcup_{v:\, u(v)\neq 0} P_v \Subset D. \tag{11}$$

This is the only localization needed in the analytic estimates. In the application, the support is contained in $\mathcal{V}[U]$ and (11) follows from $U \Subset D$ once the mesh is sufficiently small. Under (11), the piecewise-constant extension used below vanishes in a neighborhood of ∂D, so its zero extension to $\mathbb{R}^d$ creates no ambient-boundary variation.

Fix an edge $e = \{v, w\}$. Let $h_{v,e}$ and $h_{w,e}$ be the Euclidean distances from v and w, respectively, to the hyperplane containing H_e. The marked points lie on opposite sides of this hyperplane and $w - v$ is normal to it, so

$$\|e\| = h_{v,e} + h_{w,e}. \tag{12}$$

The pyramids $\mathrm{Conv}(v, H_e)$, as e ranges over the edges incident to v, partition P_v up to sets of d-dimensional measure zero. Therefore

$$\sum_{e\ni v} b(e) h_{v,e} = d m_v. \tag{13}$$

This is the same pyramid decomposition used in Bou-Rabee and Gwynne [2, Lemma 2.4].

Define an auxiliary vertex measure by

$$\nu_v := \frac{1}{2}\sum_{e\ni v} a(e)\|e\|^2 = \frac{1}{2}\sum_{e\ni v} b(e)\|e\|. \tag{14}$$

Lemma 2.1 (Comparison of ν with tile volume)**.** *For every finitely supported $u : \mathcal{V} \to \mathbb{R}$,*

$$\sum_{v\in\mathcal{V}} \nu_v u(v)^2 \le \frac{3d}{2}\|u\|_{2,m}^2 + 4\epsilon^2 \mathcal{D}(u). \tag{15}$$

Proof. By (12), (13), and (14),

$$\begin{aligned}\sum_v \nu_v u(v)^2 &= \frac{1}{2}\sum_v \sum_{e=\{v,w\}\ni v} b(e)(h_{v,e} + h_{w,e})u(v)^2 \\ &= \frac{d}{2}\sum_v m_v u(v)^2 + \frac{1}{2}\sum_v \sum_{e=\{v,w\}\ni v} b(e) h_{w,e} u(v)^2.\end{aligned}$$

For each oriented incidence (v,e) with $e=\{v,w\}$,

$$u(v)^2 \le 2u(w)^2 + 2(u(v)-u(w))^2.$$

Substitution into the second double sum gives

$$\sum_v \nu_v u(v)^2 \le \frac{d}{2}\|u\|_{2,m}^2 + \sum_v \sum_{e=\{v,w\}\ni v} b(e)h_{w,e}u(w)^2 + \sum_v \sum_{e=\{v,w\}\ni v} b(e)h_{w,e}(u(v)-u(w))^2.$$

In the first double sum on the right, collect terms according to w and use (13); it equals $d\|u\|_{2,m}^2$. In the second double sum, the two orientations of each edge combine, by (12), to give

$$\sum_{e=\{v,w\}} b(e)\|e\|(u(v)-u(w))^2 = \sum_{e=\{v,w\}} a(e)\|e\|^2(u(v)-u(w))^2.$$

If $y\in H_e$, then

$$\|e\| = \|w-v\| \le \|w-y\| + \|y-v\| \le \operatorname{diam}(P_w) + \operatorname{diam}(P_v) \le 2\epsilon.$$

The last display is therefore at most $4\epsilon^2\mathcal{D}(u)$, which proves (15). □

Set

$$q := \frac{2d}{d-1}. \tag{16}$$

Lemma 2.2 (BV–Sobolev inequality)**.** *There is a constant $C_d<\infty$ such that every finitely supported $u:\mathcal{V}\to\mathbb{R}$ satisfies*

$$\|u\|_{q,m}^2 \le C_d\mathcal{D}(u)^{1/2}\big(\|u\|_{2,m}^2 + \epsilon^2\mathcal{D}(u)\big)^{1/2}. \tag{17}$$

Proof. Extend u by zero to all vertices and define the piecewise-constant function

$$F(y) := u(v)^2 \qquad \text{for a.e. } y\in P_v. \tag{18}$$

Since u has finite support, F has compact support in D and its zero extension belongs to $BV(\mathbb{R}^d)$. We first record its total variation. Cellwise integration by parts against a vector field $\varphi\in C_c^1(\mathbb{R}^d;\mathbb{R}^d)$ gives

$$\int_{\mathbb{R}^d} F\,\operatorname{div}\varphi\,\mathrm{d}y = \sum_v u(v)^2\int_{P_v}\operatorname{div}\varphi\,\mathrm{d}y = \sum_{e=\{v,w\}}(u(v)^2-u(w)^2)\int_{H_e}\varphi\cdot n_{v,w}\,\mathrm{d}\mathcal{H}^{d-1},$$

where $n_{v,w}$ is the unit normal to H_e pointing from P_v into P_w. There is no contribution from ∂D because F vanishes in a neighborhood of it. Taking the supremum over $\|\varphi\|_\infty\le 1$ yields

$$|DF|(\mathbb{R}^d) \le \sum_{e=\{v,w\}} b(e)|u(v)^2-u(w)^2|. \tag{19}$$

In fact equality holds, but the upper bound is all that is needed.

The Euclidean BV–Sobolev inequality [1, Theorem 3.47] states that

$$\|F\|_{L^{d/(d-1)}(\mathbb{R}^d)} \le C_d|DF|(\mathbb{R}^d). \tag{20}$$

The left side equals $\|u\|_{q,m}^2$. For the right side, use

$$|u(v)^2 - u(w)^2| = |u(v) - u(w)|\,|u(v) + u(w)|$$

and Cauchy–Schwarz to obtain

$$\sum_e b(e)|u(v)^2 - u(w)^2| \le \left(\sum_e a(e)(u(v) - u(w))^2\right)^{1/2} \times \left(\sum_e b(e)\|e\|(|u(v)| + |u(w)|)^2\right)^{1/2}.$$

Since $(|s| + |t|)^2 \le 2s^2 + 2t^2$ and $\sum_{e\ni v} b(e)\|e\| = 2\nu_v$, the second squared factor is at most

$$4\sum_v \nu_v u(v)^2.$$

Consequently, Lemma 2.1 gives

$$|DF|(\mathbb{R}^d) \le C_d \mathcal{D}(u)^{1/2}\big(\|u\|_{2,m}^2 + \epsilon^2 \mathcal{D}(u)\big)^{1/2}.$$

Combining this estimate with (20) proves (17). □

Proposition 2.3 (Defective Nash and support inequalities)**.** *There is a constant $C_d < \infty$ such that every finitely supported $u : \mathcal{V} \to \mathbb{R}$ satisfies*

$$\|u\|_{2,m}^{4+4/d} \le C_d \|u\|_{1,m}^{4/d} \mathcal{D}(u)\big(\|u\|_{2,m}^2 + \epsilon^2 \mathcal{D}(u)\big). \tag{21}$$

If

$$V_u := \sum_{v : u(v) \neq 0} m_v, \tag{22}$$

then

$$\|u\|_{2,m}^2 \le C_d\big(V_u^{2/d} + \epsilon V_u^{1/d}\big)\mathcal{D}(u). \tag{23}$$

Proof. The log-convexity interpolation inequality for L^p norms, in the form stated in Folland [4, Section 6.5], gives

$$\|u\|_{2,m} \le \|u\|_{1,m}^{1/(d+1)} \|u\|_{q,m}^{d/(d+1)}. \tag{24}$$

Indeed,

$$\frac{1}{2} = \frac{1}{d+1} \cdot 1 + \frac{d}{d+1} \cdot \frac{1}{q}, \qquad q = \frac{2d}{d-1}.$$

Combining (24) with (17), and then raising both sides to the power $2(d+1)/d$, gives (21).

For the support estimate, Hölder's inequality gives

$$\|u\|_{2,m}^2 \le V_u^{1/d} \|u\|_{q,m}^2. \tag{25}$$

Let $X := \|u\|_{2,m}^2$ and $Y := \mathcal{D}(u)$. From (25) and (17),

$$X^2 \le C_d V_u^{2/d} Y(X + \epsilon^2 Y). \tag{26}$$

If $Y = 0$, then (26) gives $X = 0$. Suppose $Y > 0$ and set $R := X/Y$. Dividing (26) by Y^2 gives

$$R^2 \le C_d V_u^{2/d}(R + \epsilon^2).$$

The positive root of the corresponding quadratic is bounded by

$$C_d V_u^{2/d} + C_d \epsilon V_u^{1/d}.$$

Since $X = RY$, this proves (23). □

3. Heat-kernel smoothing in tile-volume time

To avoid any issue concerning explosion of a process on an infinite graph, all processes below are first defined on a finite vertex set. Let $W \subset \mathcal{V}$ be finite and assume that every neighbor of a vertex in W lies in a finite set $\overline{W}$. Declare the vertices of $\overline{W} \setminus W$ absorbing. On W, define the continuous-time generator

$$(\mathcal{L}f)(v) := \frac{1}{m_v}\Delta_a^{\mathcal{G}} f(v) = \frac{1}{m_v}\sum_{w\sim v} a(v,w)(f(w) - f(v)). \tag{27}$$

Thus the jump rate from v to w is $a(v,w)/m_v$. The total holding time at v is exponential with mean

$$\frac{m_v}{a(v)}, \qquad a(v) := \sum_{w\sim v} a(v,w),$$

and, conditional on jumping, the next vertex is w with probability $a(v,w)/a(v)$. The measure m is reversible for the process.

Let $S \subset W$ and let

$$\tau_S := \inf\{t \geq 0 : \widetilde{X}_t \notin S\}.$$

Write $p_t^S(x,y)$ for the transition density of the process killed on exiting S, with respect to m:

$$\mathbb{P}_x[\widetilde{X}_t = y,\ t < \tau_S] = m_y p_t^S(x,y). \tag{28}$$

Since the killed state space is finite, its semigroup is a differentiable self-adjoint contraction semigroup on $L^2(S,m)$. In particular,

$$p_t^S(x,y) = p_t^S(y,x), \qquad p_{s+t}^S(x,y) = \sum_{z\in S} m_z p_s^S(x,z) p_t^S(z,y). \tag{29}$$

These are the standard finite-state continuous-time Markov-chain identities; see, for example, Norris [5, Chapter 2].

Proposition 3.1 (On-diagonal and survival bounds)**.** *Suppose that $S \subset W$ is finite and that $\bigcup_{v\in S} P_v \Subset D$. There is a constant $C_d < \infty$, independent of S, such that, for every $x \in S$ and every $t > 0$,*

$$p_{2t}^S(x,x) \leq C_d\big(t^{-d/2} + \epsilon^d t^{-d}\big). \tag{30}$$

Consequently, whenever $t \geq \epsilon^2$,

$$\mathbb{P}_x(\tau_S > t) \leq C_d m(S)^{1/2} t^{-d/4}, \qquad m(S) := \sum_{v\in S} m_v. \tag{31}$$

Proof. Fix $x \in S$ and define

$$u_t(y) := p_t^S(x,y), \qquad X(t) := \|u_t\|_{2,m}^2.$$

By symmetry and the semigroup identity in (29),

$$X(t) = \sum_{y\in S} m_y p_t^S(x,y)^2 = p_{2t}^S(x,x). \tag{32}$$

Let $\mathcal{L}_S$ denote the killed generator, meaning that u_t is extended by zero outside S before (27) is applied. Self-adjointness and the weighted summation-by-parts identity give

$$X'(t) = 2\langle u_t, \mathcal{L}_S u_t\rangle_{L^2(m)} = -2\mathcal{D}(u_t). \tag{33}$$

For completeness, the last identity follows by pairing the two orientations of every edge:

$$-\sum_v m_v u(v)(\mathcal{L}_S u)(v) = \sum_{\{v,w\}\in\mathcal{E}} a(v,w)(u(v)-u(w))^2.$$

Also,

$$\|u_t\|_{1,m} = \sum_{y\in S} m_y p_t^S(x,y) = \mathbb{P}_x(t<\tau_S) \le 1. \tag{34}$$

Set

$$Q(t) := \mathcal{D}(u_t) = -\frac{1}{2}X'(t). \tag{35}$$

Applying (21) to u_t and using (34) gives

$$X(t)^{2+2/d} \le C_d Q(t)(X(t)+\epsilon^2 Q(t)). \tag{36}$$

We next extract a scalar differential inequality from (36). If $X(t)\ge \epsilon^2 Q(t)$, then

$$X(t)+\epsilon^2 Q(t) \le 2X(t),$$

so (36) gives

$$Q(t) \ge c_d X(t)^{1+2/d}. \tag{37}$$

If $X(t) < \epsilon^2 Q(t)$, then

$$X(t)+\epsilon^2 Q(t) < 2\epsilon^2 Q(t),$$

and (36) gives

$$Q(t) \ge c_d \epsilon^{-1} X(t)^{1+1/d}. \tag{38}$$

Using (35), the two cases imply

$$-X'(t) \ge c_d \min\left\{X(t)^{1+2/d},\, \epsilon^{-1}X(t)^{1+1/d}\right\}. \tag{39}$$

The terms inside the minimum agree exactly at $X=\epsilon^{-d}$.

Define the first threshold-crossing time

$$\sigma := \inf\{t\ge 0 : X(t)\le \epsilon^{-d}\}. \tag{40}$$

The function X is continuous and nonincreasing by (33). Thus $X(t)>\epsilon^{-d}$ for $0\le t<\sigma$, while $X(t)\le\epsilon^{-d}$ for $t\ge\sigma$. If $0<t<\sigma$, then the smaller term in (39) is $\epsilon^{-1}X(t)^{1+1/d}$, and hence

$$\frac{\mathrm{d}}{\mathrm{d}t}X(t)^{-1/d} = \frac{1}{d}X(t)^{-1-1/d}(-X'(t)) \ge c_d\epsilon^{-1}. \tag{41}$$

Integrating from 0 to $t<\sigma$ gives

$$X(t)^{-1/d} \ge X(0)^{-1/d} + c_d\epsilon^{-1}t \ge c_d\epsilon^{-1}t,$$

so

$$X(t) \le C_d \epsilon^d t^{-d}, \qquad 0<t<\sigma. \tag{42}$$

This estimate also shows that

$$\sigma \le C_d\epsilon^2. \tag{43}$$

Indeed, choose K_d so large that $C_d K_d^{-d}\le 1$. If $\sigma > K_d\epsilon^2$, then (42) at $t=K_d\epsilon^2$ would give $X(t)\le\epsilon^{-d}$, contradicting the definition of σ.

For $t\ge\sigma$, the smaller term in (39) is $X(t)^{1+2/d}$. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}X(t)^{-2/d} = \frac{2}{d}X(t)^{-1-2/d}(-X'(t)) \ge c_d. \tag{44}$$

Integrating from σ to t and using $X(\sigma) \le \epsilon^{-d}$ gives

$$X(t) \le C_d\big(\epsilon^2 + t - \sigma\big)^{-d/2}, \qquad t \ge \sigma. \tag{45}$$

If $t \ge 2\sigma$, then $t - \sigma \ge t/2$, so (45) gives $X(t) \le C_d t^{-d/2}$. If $\sigma \le t < 2\sigma$, then monotonicity and (43) give

$$X(t) \le \epsilon^{-d} \le C_d \epsilon^d t^{-d}.$$

Together with (42), these estimates prove

$$X(t) \le C_d(t^{-d/2} + \epsilon^d t^{-d})$$

for every $t > 0$. The on-diagonal bound (30) now follows from (32).

Finally, Cauchy–Schwarz and (32) imply

$$\begin{aligned}\mathbb{P}_x(\tau_S > t) &= \sum_{y\in S} m_y p_t^S(x,y) \\ &\le m(S)^{1/2}\left(\sum_{y\in S} m_y p_t^S(x,y)^2\right)^{1/2} \\ &= m(S)^{1/2} p_{2t}^S(x,x)^{1/2}.\end{aligned}$$

When $t \ge \epsilon^2$,

$$\epsilon^d t^{-d} = t^{-d/2}\left(\frac{\epsilon^2}{t}\right)^{d/2} \le t^{-d/2}.$$

Applying (30) proves (31). □

4. The local tile-volume clock

The next construction supplies a Euclidean lower bound on the auxiliary clock. Although the tiling is not assumed to be given by a power diagram, the orthogonality relation implies that compatible power weights exist locally.

Lemma 4.1 (Local power potential)**.** *Let $B_R(z) \Subset D$. Define a piecewise-affine vector field on the ball by*

$$Z(y) := 2(y - v), \qquad y \in \operatorname{int} P_v \cap B_R(z). \tag{46}$$

Then there is a function $\psi \in W^{1,\infty}(B_R(z))$ such that $\nabla\psi = Z$ almost everywhere. For every cell whose interior meets $B_R(z)$, there is a constant $\omega_v \in \mathbb{R}$ such that

$$\psi(y) = \|y - v\|^2 - \omega_v, \qquad y \in P_v \cap B_R(z). \tag{47}$$

Moreover,

$$\|\nabla\psi\|_{L^\infty(B_R(z))} \le 2\epsilon. \tag{48}$$

Proof. We first prove that Z is curl-free in the sense of distributions. Inside each cell, its derivative is the symmetric matrix $2I$. Across a facet $H_e = P_v \cap P_w$,

$$Z|_{P_w} - Z|_{P_v} = 2(v - w),$$

which is parallel to the normal of H_e. Therefore the tangential component of Z has the same trace from the two sides. Cellwise integration by parts against a compactly supported smooth test field shows that all interface contributions to $\partial_i Z_j - \partial_j Z_i$ vanish. Hence

$$\partial_i Z_j - \partial_j Z_i = 0 \quad \text{in } \mathcal{D}'(B_R(z)) \quad \text{for all } i, j. \tag{49}$$

For completeness, we construct the potential from (49). Fix $0 < \rho < R$. Choose $\chi \in C_c^\infty(B_R(z))$ which equals one on a neighborhood of $\overline{B_\rho(z)}$, extend χZ by zero to $\mathbb{R}^d$, and let η_δ be a standard mollifier. If δ is smaller than the distance from $B_\rho(z)$ to the set where $\chi \neq 1$, then

$$Z_\delta := (\chi Z) * \eta_\delta$$

is smooth on $B_\rho(z)$. On this ball its distributional curl is the convolution of the distributional curl of Z with η_δ, so it is zero by (49). Since the ball is convex, the function

$$\psi_\delta(x) := \int_0^1 Z_\delta(z + s(x-z)) \cdot (x-z)\, \mathrm{d}s$$

satisfies $\nabla \psi_\delta = Z_\delta$ and $\psi_\delta(z) = 0$. Furthermore,

$$\|\nabla \psi_\delta\|_{L^\infty(B_\rho(z))} \leq \|Z\|_{L^\infty(B_R(z))}.$$

After passing to a subsequence, the Arzelà–Ascoli theorem gives uniform convergence on compact subballs to a Lipschitz function ψ, and $Z_\delta \to Z$ in L^1_{loc}. It follows distributionally that $\nabla \psi = Z$. A diagonal argument in ρ gives a single function on $B_R(z)$.

For almost every $y \in P_v \cap B_R(z)$,

$$\nabla\big(\psi(y) - \|y - v\|^2\big) = 0.$$

The set $P_v \cap B_R(z)$ is convex and hence connected, so this difference is constant there. Taking the continuous representative of the Lipschitz function ψ gives (47). Finally, if $y \in P_v$, then $\|y - v\| \leq \operatorname{diam}(P_v) \leq \epsilon$, so (46) gives (48). □

Lemma 4.2 (Exact clock identity)**.** *Suppose that $B_{r+3\epsilon}(z) \Subset D$, and choose the weights (ω_v) from Lemma 4.1 on this ball. Define*

$$\Phi_z(v) := \|v - z\|^2 - \omega_v. \tag{50}$$

Then, for every $v \in \mathcal{V} \cap B_r(z)$,

$$\mathcal{L}\Phi_z(v) = 2d. \tag{51}$$

Moreover, if $v, w \in \mathcal{V} \cap B_{r+2\epsilon}(z)$, then

$$|\omega_v - \omega_w| \leq 2\epsilon \|v - w\|. \tag{52}$$

Proof. Let $e = \{v, w\}$ and choose $y \in H_e$ in the ball on which (47) is valid. Continuity of ψ gives

$$\|y - v\|^2 - \omega_v = \|y - w\|^2 - \omega_w.$$

A direct expansion therefore yields

$$\Phi_z(w) - \Phi_z(v) = 2(w - v) \cdot (y - z). \tag{53}$$

Now suppose $v \in B_r(z)$. Every point of P_v belongs to $B_{r+\epsilon}(z)$, and every neighbor w of v belongs to $B_{r+2\epsilon}(z)$. Thus (53) is available on every facet of P_v. Let c_e be the centroid of H_e and let

$$n_{v,w} := \frac{w - v}{\|w - v\|}$$

be the unit normal pointing out of P_v across H_e. Since the normal component of $y - z$ is constant on H_e, (1) and (53) imply

$$\begin{aligned}\Delta_a^{\mathcal{G}}\Phi_z(v) &= 2\sum_{w\sim v} b(e)n_{v,w}\cdot(c_e - z)\\ &= 2\int_{\partial P_v}(y-z)\cdot n(y)\,\mathrm{d}\mathcal{H}^{d-1}(y)\\ &= 2d\,\mathrm{vol}_d(P_v) = 2dm_v.\end{aligned}$$

The penultimate equality is the divergence theorem. Dividing by m_v proves (51).

For the second assertion, (47) evaluated at the marked points gives $\psi(v) = -\omega_v$ and $\psi(w) = -\omega_w$. The segment from v to w lies in $B_{r+2\epsilon}(z)$ because this ball is convex. Hence (48) and the fundamental theorem for Lipschitz functions on line segments give

$$|\omega_v - \omega_w| = |\psi(v) - \psi(w)| \le 2\epsilon\|v - w\|.$$

□

Proposition 4.3 (A fixed ball cannot be crossed instantly)**.** *Let $\widetilde{X}$ be the finite-state process with generator* (27)*, started from a vertex z. Let*

$$\tau_{B_r} := \inf\{t \ge 0 : \widetilde{X}_t \notin B_r(z)\}.$$

Assume that $B_{r+3\epsilon}(z) \Subset D$, and choose the local weights (ω_v) and the function Φ_z as in Lemma 4.2. Let τ be any stopping time. Then

$$\mathbb{P}_z(\tau_{B_r} < t \wedge \tau) \le \frac{2dt + 2\epsilon(r+2\epsilon)}{r^2}. \tag{54}$$

Proof. Extend Φ_z arbitrarily to the rest of the finite state space. This does not change (51), since every neighbor of a vertex in $B_r(z)$ lies in $B_{r+2\epsilon}(z)$, where the original definition of Φ_z is available. Set $T := t \wedge \tau_{B_r} \wedge \tau$. Since $T \le t$, the finite-state Dynkin formula [5, Chapter 2], applied to the extended function and (51), gives

$$\mathbb{E}_z[\Phi_z(\widetilde{X}_T)] = \Phi_z(z) + 2d\mathbb{E}_z[T]. \tag{55}$$

Since $\Phi_z(z) = -\omega_z$, this identity is equivalent to

$$\mathbb{E}_z\|\widetilde{X}_T - z\|^2 = 2d\mathbb{E}_z[T] + \mathbb{E}_z[\omega_{\widetilde{X}_T} - \omega_z]. \tag{56}$$

Every jump has length at most 2ϵ. Therefore $\widetilde{X}_T \in B_{r+2\epsilon}(z)$, and (52) gives

$$|\omega_{\widetilde{X}_T} - \omega_z| \le 2\epsilon\|\widetilde{X}_T - z\| \le 2\epsilon(r+2\epsilon).$$

Using $T \le t$ in (56), we obtain

$$\mathbb{E}_z\|\widetilde{X}_T - z\|^2 \le 2dt + 2\epsilon(r+2\epsilon). \tag{57}$$

On the event $\{\tau_{B_r} < t \wedge \tau\}$, one has $T = \tau_{B_r}$ and $\|\widetilde{X}_T - z\| \ge r$. Hence

$$r^2\mathbb{P}_z(\tau_{B_r} < t \wedge \tau) \le \mathbb{E}_z\|\widetilde{X}_T - z\|^2.$$

Combining this with (57) proves (54). □

Remark 4.4. The normalization in (27) approximates Δ, rather than $\frac{1}{2}\Delta$. This is why the corrected squared displacement in (56) grows at rate $2d$. Replacing the rates by $a(v,w)/(2m_v)$ changes only the auxiliary time scale.

5. Energy and weighted L^2 error

The required estimate is a standard finite-volume consistency bound; see Eymard et al. [3, Theorem 9.3] for the general finite-volume framework. We use the orthogonal-tiling form proved by Bou-Rabee and Gwynne [2, Proposition 2.6], and no part of that energy estimate is reproved here. Fix a bounded open set K such that

$$\overline{U} \subset K \Subset D, \tag{58}$$

and suppose that h_C is C^2 and harmonic on a neighborhood of $\overline{K}$. Put

$$\delta_K := \operatorname{dist}(\overline{U}, K^c) > 0. \tag{59}$$

If $v \in \mathcal{V}[U]$, then $P_v \subset B_\epsilon(U)$. If $w \in \partial\mathcal{V}[U]$ is adjacent to $v \in \mathcal{V}[U]$, then $\|w - v\| \le 2\epsilon$ and hence $P_w \subset B_{3\epsilon}(U)$. Consequently,

$$3\epsilon < \delta_K \quad \Longrightarrow \quad \widehat{U} \subset K. \tag{60}$$

For an edge $e = \{v, w\}$, set

$$Q_e := \operatorname{Conv}(v, H_e) \cup \operatorname{Conv}(w, H_e). \tag{61}$$

By Bou-Rabee and Gwynne [2, Lemma 2.4],

$$\|e\| b(e) = d \operatorname{vol}_d(Q_e). \tag{62}$$

The inclusion $\bigcup_{e\in\overline{\mathcal{E}[U]}} Q_e \subset \widehat{U}$ is Bou-Rabee and Gwynne [2, Lemma 2.5]. We need the corresponding bound for the sum of volumes. For a fixed v, the interiors of the pyramids $\operatorname{Conv}(v, H_e)$, $e \ni v$, are pairwise disjoint and their union is P_v up to a null set. Pyramids belonging to distinct cells also have disjoint interiors. Therefore all oriented pyramids which occur in the sets Q_e, $e \in \overline{\mathcal{E}[U]}$, have pairwise disjoint interiors, and

$$\sum_{e\in\overline{\mathcal{E}[U]}} \operatorname{vol}_d(Q_e) \le \sum_{v\in\overline{\mathcal{V}[U]}} \operatorname{vol}_d(P_v) = \operatorname{vol}_d(\widehat{U}) \le \operatorname{vol}_d(K), \tag{63}$$

where the last inequality uses (60).

Lemma 5.1 (Energy and weighted L^2 error)**.** *Let h_D solve* (7)*, let*

$$f := h_D - h_C \quad \text{on } \overline{\mathcal{V}[U]},$$

and extend f by zero to all of $\mathcal{V}$. Set

$$M := \|\nabla^2 h_C\|_{L^\infty(K)}.$$

Then, for all sufficiently small ϵ,

$$\mathcal{D}(f) + \|f\|_{2,m}^2 \le C_{d,K} M^2 \epsilon^2. \tag{64}$$

Proof. The boundary values in (7) imply that $f = 0$ on $\partial\mathcal{V}[U]$, so its global energy equals $\mathcal{D}(f, U)$. The finite-volume consistency estimate [2, Proposition 2.6] gives

$$\mathcal{D}(f) \le 9M^2\epsilon^2 \sum_{e\in\overline{\mathcal{E}[U]}} d \operatorname{vol}_d(Q_e). \tag{65}$$

Combining (65) with (63) gives

$$\mathcal{D}(f) \le 9d \operatorname{vol}_d(K) M^2 \epsilon^2. \tag{66}$$

The support of f is contained in $\mathcal{V}[U]$, so

$$V_f = \sum_{v:f(v)\neq 0} m_v \leq \mathrm{vol}_d(\widehat{U}) \leq \mathrm{vol}_d(K).$$

Applying the support inequality (23) and then (66), and using $\epsilon \leq 1$, gives

$$\|f\|_{2,m}^2 \leq C_{d,K}\mathcal{D}(f) \leq C_{d,K}M^2\epsilon^2.$$

This proves (64). □

6. Direct uniform estimate

Proof of Theorem 1.1. We prove the quantitative estimate (10) directly. In particular, we do not argue by contradiction or pass to a subsequence.

Choose $r_0 > 0$ so small that

$$B_{4r_0}(U) \Subset K. \tag{67}$$

Let

$$H := \|h_C\|_{L^\infty(K)}, \qquad L := \|\nabla h_C\|_{L^\infty(K)}, \tag{68}$$

$$M := \|\nabla^2 h_C\|_{L^\infty(K)}, \qquad H_* := 1 + H. \tag{69}$$

The finite-network maximum principle (8), applied to h_D and then to $-h_D$, gives

$$\max_{v\in\overline{\mathcal{V}[U]}} |h_D(v)| \leq H. \tag{70}$$

Let

$$A := \max_{v\in\mathcal{V}[U]} (h_D(v) - h_C(v)). \tag{71}$$

If $A \leq 0$, there is nothing to prove for the positive part of the error. If $H = 0$, then (70) gives $h_D = 0$ and, since $h_C = 0$ on K, the entire error vanishes. Hence we may suppose that $A > 0$ and $H > 0$, and choose $z \in \mathcal{V}[U]$ such that

$$h_D(z) - h_C(z) = A, \tag{72}$$

and note from (70) that

$$A \leq 2H. \tag{73}$$

Set

$$\Lambda := 64\left(1 + L + \frac{H}{r_0}\right), \qquad r := \frac{A}{\Lambda}. \tag{74}$$

Then (73) implies $r \leq r_0/32$, and

$$Lr \leq \frac{A}{64}. \tag{75}$$

We shall separate a harmless small-error regime. Let

$$c_0 := \frac{3}{32H_*}. \tag{76}$$

Choose $K_0 \geq 1$, depending only on d, H, L, r_0, so large that

$$K_0 \geq 128L, \qquad K_0^2 \geq \frac{16\Lambda}{c_0}, \qquad K_0^3 \geq \frac{32\Lambda^2}{c_0}. \tag{77}$$

Set

$$\kappa := \frac{c_0}{8d\Lambda^2}. \tag{78}$$

Choose $\epsilon_0 > 0$ so that

$$\epsilon_0 \le \min\left\{1, r_0, \frac{\delta_K}{3}, (\kappa K_0^3)^2\right\}. \tag{79}$$

We prove the asserted quantitative estimate for $0 < \epsilon \le \epsilon_0$. If

$$A \le K_0 \epsilon^{1/2}, \tag{80}$$

then, since $d \ge 2$ and $0 < \epsilon \le 1$,

$$\frac{1}{2} > \beta_d = \frac{4}{3d+8} \quad \Longrightarrow \quad A \le K_0 \epsilon^{\beta_d}.$$

It remains to treat the case

$$A > K_0 \epsilon^{1/2}. \tag{81}$$

Under (81), the first condition in (77), together with $\epsilon \le 1$, implies

$$L(r+2\epsilon) \le \frac{A}{64} + 2L\epsilon < \frac{A}{64} + \frac{2L}{K_0} A\epsilon^{1/2} \le \frac{A}{32} < \frac{A}{8}.$$

Thus

$$L(r+2\epsilon) \le \frac{A}{8}. \tag{82}$$

Moreover, $r \le r_0/32$ and $\epsilon \le r_0$, so (67) gives $B_{r+3\epsilon}(z) \Subset K$.

Define the superlevel set

$$S := \{v \in \mathcal{V}[U] : h_D(v) - h_C(v) > A/2\}. \tag{83}$$

By Lemma 5.1 and Chebyshev's inequality,

$$m(S) \le \frac{4}{A^2} \|h_D - h_C\|_{2,m}^2 \le C_{d,K} M^2 \epsilon^2 A^{-2}. \tag{84}$$

Run the finite-state process (27) on $\overline{\mathcal{V}[U]}$, with $\partial\mathcal{V}[U]$ absorbing, and start it from z. Let τ_U be its first hitting time of $\partial\mathcal{V}[U]$. Let τ_S and τ_{B_r} be the first exit times from S and $B_r(z)$, respectively, and set

$$E := \{\tau_{B_r} < \tau_S\}, \qquad p := \mathbb{P}_z(E), \qquad \sigma := \tau_{B_r} \wedge \tau_S. \tag{85}$$

Since $S \subset \mathcal{V}[U]$, hitting $\partial\mathcal{V}[U]$ necessarily exits S, so

$$\tau_S \le \tau_U \qquad \text{and hence} \qquad \sigma \le \tau_U. \tag{86}$$

The process

$$h_D(\widetilde{X}_{t\wedge\tau_U})$$

is a bounded martingale: this follows from $\mathcal{L}h_D = m^{-1}\Delta_a^{\mathcal{G}} h_D = 0$ on $\mathcal{V}[U]$ and the finite-state Dynkin formula. For complete justification of the stopping step, apply optional stopping first at $\sigma \wedge N$. Each connected component of the finite set S has a positive-conductance edge to its complement, so the finite-state chain exits S almost surely; in particular, $\sigma \le \tau_S < \infty$ almost surely. Since h_D is bounded, bounded convergence as $N \to \infty$ yields

$$h_D(z) = \mathbb{E}_z[h_D(\widetilde{X}_\sigma)]. \tag{87}$$

On E^c, the terminal vertex lies outside S and is at distance at most $r + 2\epsilon$ from z. Consequently,

$$h_D(\widetilde{X}_\sigma) \le h_C(z) + L(r+2\epsilon) + A/2 \qquad \text{on } E^c. \tag{88}$$

On E, the maximum principle gives $h_D(\widetilde{X}_\sigma) \le H$. Combining (72), (87), and (88), and then using (82), gives

$$\begin{aligned} h_C(z) + A &\le h_C(z) + L(r+2\epsilon) + A/2 \\ &\quad + p\big(2H + L(r+2\epsilon) + A/2\big), \end{aligned}$$

Writing $B := L(r+2\epsilon)$, the preceding inequality is

$$\frac{A}{2} - B \le p\left(2H + B + \frac{A}{2}\right).$$

By (82), the left side is at least $3A/8$. By (73),

$$2H + B + \frac{A}{2} \le 2H + \frac{5A}{8} \le 2H + \frac{5H}{4} \le 4H_*.$$

Consequently,

$$p \ge \frac{3A}{32H_*} = c_0 A. \tag{89}$$

We now give an upper bound for the same probability. Set

$$t := \kappa A^3. \tag{90}$$

By (81) and (79),

$$t > \kappa K_0^3 \epsilon^{3/2} \ge \epsilon^2.$$

Proposition 4.3, with killing time τ_U, gives

$$\begin{aligned} \mathbb{P}_z(\tau_{B_r} < t \wedge \tau_U) &\le \frac{2dt + 2\epsilon(r+2\epsilon)}{r^2} \\ &= 2d\kappa\Lambda^2 A + \frac{2\Lambda\epsilon}{A} + \frac{4\Lambda^2\epsilon^2}{A^2}. \end{aligned} \tag{91}$$

The definition of κ makes the first term equal to $c_0 A/4$. Since $A^2 > K_0^2\epsilon$, the second term satisfies

$$\frac{2\Lambda\epsilon}{A} < \frac{2\Lambda}{K_0^2} A \le \frac{c_0 A}{8}.$$

Also, $A^3 > K_0^3\epsilon^{3/2} \ge K_0^3\epsilon^2$ because $\epsilon \le 1$, so

$$\frac{4\Lambda^2\epsilon^2}{A^2} < \frac{4\Lambda^2}{K_0^3} A \le \frac{c_0 A}{8}.$$

Hence, by (89),

$$\mathbb{P}_z(\tau_{B_r} < t \wedge \tau_U) \le \frac{c_0 A}{2} \le \frac{p}{2}. \tag{92}$$

If E occurs and $\tau_{B_r} \ge t$, then necessarily $\tau_S > t$. If E occurs and $\tau_{B_r} < t$, then $\tau_{B_r} < t \wedge \tau_U$, because $\tau_{B_r} < \tau_S \le \tau_U$ by (86). Therefore

$$p \le \mathbb{P}_z(\tau_S > t) + \mathbb{P}_z(\tau_{B_r} < t \wedge \tau_U). \tag{93}$$

Combining (92) and (93) yields

$$\frac{p}{2} \le \mathbb{P}_z(\tau_S > t). \tag{94}$$

By (31), (84), and (90),

$$\begin{aligned} \mathbb{P}_z(\tau_S > t) &\le C_d m(S)^{1/2} t^{-d/4} \\ &\le C_{d,K,h_C} \epsilon A^{-1} (A^3)^{-d/4} = C_{d,K,h_C} \epsilon A^{-1-3d/4}. \end{aligned} \tag{95}$$

Using (89), (94), and (95), we obtain

$$cA \le C\epsilon A^{-1-3d/4}.$$

Equivalently,

$$A^{2+3d/4} \le C\epsilon. \tag{96}$$

Since $(2+3d/4)^{-1} = 4/(3d+8)$, (96) gives

$$\max_{v\in\mathcal{V}[U]} (h_D(v) - h_C(v)) \le C\epsilon^{\beta_d}. \tag{97}$$

This includes both the small-error regime (80) and the large-error regime (81).

Apply the same argument to the continuum harmonic function $-h_C$ and its discrete Dirichlet extension $-h_D$. The quantities H, L, M are unchanged. We therefore also have

$$\max_{v\in\mathcal{V}[U]} (h_C(v) - h_D(v)) \le C\epsilon^{\beta_d}.$$

Together with (97), this proves (10). For the qualitative statement (9), choose a bounded open set K whose closure is contained in the neighborhood on which h_C is harmonic and which contains $\overline{U}$. This completes the proof of Theorem 1.1. □

7. Deduction of the random-walk consequences

Proof of Corollary 1.2. Let D_0 denote the domain called D in Assumption D of Bou-Rabee and Gwynne [2, Section 4], and let D_{amb} be the ambient domain tiled by the graphs. Choose $\delta > 0$ such that

$$B_{2\delta}(D_0) \Subset D_{\mathrm{amb}}.$$

For $r > 0$ and $x \in B_\delta(D_0)$, set

$$B := B_r(x) \cap B_\delta(D_0),$$

exactly as in that assumption. This is a bounded domain with $\overline{B} \Subset D_{\mathrm{amb}}$.

We first verify Bou-Rabee and Gwynne [2, Assumption D, equation (4.1)]. If $y \in B$ and $y \in P_v$, then

$$\mathrm{dist}(y, \mathcal{V}_n) \le \|y - v\| \le \mathrm{diam}(P_v) \le \epsilon_n.$$

Hence the nearest-vertex distance tends to zero uniformly on B. Next let h_C be harmonic on a neighborhood of $\overline{B}$. Applying Theorem 1.1 with $U = B$ gives Bou-Rabee and Gwynne [2, Assumption D, equation (4.2)]. Thus Assumption D is verified.

Under Assumption D, Bou-Rabee and Gwynne [2, Proposition 4.2] gives uniform convergence of harmonic measure on balls. The proof of Bou-Rabee and Gwynne [2, Theorem 4.1] then concatenates successive ball exits by the strong Markov property and yields convergence of the stopped canonical walk modulo time parametrization. Finally, the argument labeled "Proof of Theorem B assuming Hypothesis (b)" in Bou-Rabee and Gwynne [2, Section 4] transfers this random-walk convergence to the Dirichlet problem with continuous boundary data on a Lipschitz domain. These invocations use condition (1.3) and Assumption D, but not alternatives (I)–(III). □

Remark 7.1 (Why arbitrarily small cells no longer cause an obstruction). The estimate (64) allows the pointwise error to be large on a set of very small tile volume. Proposition 3.1 prevents the tile-volume-time walk from remaining too long in such a set, while Proposition 4.3 prevents it from crossing a fixed Euclidean ball in too little tile-volume time. The martingale identity (87) converts a pointwise error

of size A into a crossing probability of order A. Comparing the two probabilistic bounds yields the direct inequality (96).

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